

SE LAB 3_ Architectural Pattern Selection

Topic : Student project portfolio and Showcase App

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Architecture Selection

We chose **Layered Architecture** for the Student Project Portfolio & Showcase App.

Architectural Choice

The system follows a **Layered Architecture** with separate Presentation, Business, and Data layers. This provides a clear separation between user interaction, system operations, and data storage.

Reason 1

Layered Architecture makes the system easier to manage because functions such as project submission, judging, and voting can be handled separately within the Business Layer.

Reason 2

The architecture is suitable for the project because it provides a simple and organized structure for handling student projects, judge evaluations, public voting, and leaderboard operations.

Security Advantage

The Business Layer can handle authorization, project validation, and anti-sybil voting checks before allowing access to system data. This helps prevent unauthorized access and invalid votes.

Performance Benefit

Separating the application into layers allows requests to be processed by the appropriate component without directly accessing the database. This reduces unnecessary database access and improves the efficiency of system operations.